

Homework 3: Linear Algebra

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Due: Chapter 3, ETC: 1-2 hours, 100 points

Practice Problems: Work together in class

1. Consider the following system of equations:

$$\begin{aligned}2x_1 + x_2 &= 10 \\ x_1 + 3x_2 &= 15.\end{aligned}$$

- (a) Write the system in matrix form $\mathbf{Ax} = \mathbf{b}$.
- (b) Identify the coefficient matrix $\mathbf{A}$, the vector of unknowns $\mathbf{x}$, and the vector $\mathbf{b}$.
- (c) Solve the system by substitution or elimination.
- (d) Interpret x_1 and x_2 as quantities of two goods chosen to satisfy two linear requirements.

2. Let

$$A = \begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & -1 \\ 0 & 2 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

Compute:

- (a) $A + B$
- (b) $A - C$
- (c) $3A$
- (d) $2B - C$

3. Suppose a firm uses two inputs, labor L and capital K , to produce two goods. The matrix

$$A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$$

records input requirements, where rows represent goods and columns represent inputs. Thus the first row says that one unit of good 1 requires 2 units of labor and 1 unit of capital.

- (a) If the firm produces the output vector

$$\mathbf{q} = \begin{bmatrix} 10 \\ 5 \end{bmatrix},$$

compute $A'\mathbf{q}$.

- (b) Interpret the two entries of $A'\mathbf{q}$.
- (c) Why do we use A' rather than A in this calculation?

4. Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}.$$

- (a) Compute AB .

- (b) Compute BA .
- (c) Is $AB = BA$?
- (d) What does this example show about matrix multiplication?

5. For each matrix below, determine whether it is an identity matrix, a null matrix, a symmetric matrix, or none of these. Some matrices may satisfy more than one description. Comment on the linear independence and determinant in each case.

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & 5 \\ 5 & 3 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

- (a) Classify A .
- (b) Classify B .
- (c) Classify C .
- (d) Classify D .

Homework Problems: Submit individually

6. Consider the following simple market model:

$$\begin{aligned} Q_d &= 40 - 2P \\ Q_s &= 10 + P. \end{aligned}$$

In equilibrium, $Q_d = Q_s = Q$.

- (a) Rewrite the equilibrium conditions as a system of two linear equations in the unknowns Q and P .
- (b) Write this system in matrix form $\mathbf{Ax} = \mathbf{b}$.
- (c) Solve for the equilibrium quantity Q^* and price P^* .
- (d) Briefly interpret your answer.

7. Let

$$A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}.$$

- (a) Compute $|A|$.
- (b) Does A^{-1} exist?
- (c) If the inverse exists, find A^{-1} .
- (d) Use your answer to solve

$$\mathbf{Ax} = \begin{bmatrix} 14 \\ 8 \end{bmatrix}.$$

8. Evaluate the determinants of the following matrices. Then state whether each matrix is singular or nonsingular.

- (a)

$$A = \begin{bmatrix} 3 & 1 \\ 6 & 2 \end{bmatrix}$$

- (b)

$$B = \begin{bmatrix} 2 & 5 \\ 1 & 4 \end{bmatrix}$$

(c)

$$C = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 2 & 0 & 1 \end{bmatrix}$$

9. Consider the input-output system

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 18 \\ 21 \end{bmatrix}.$$

Interpret x_1 and x_2 as production levels of two sectors.

- (a) Use Cramer's Rule to solve for x_1 and x_2 .
- (b) Check your answer by substituting it back into the original system.
- (c) Is the coefficient matrix nonsingular?
- (d) What does nonsingularity imply about the number of solutions?

10. **Slightly harder problem.** Suppose a consumer's total spending on two goods must satisfy two linear restrictions:

$$\begin{aligned} 2x_1 + 4x_2 &= 20 \\ x_1 + 2x_2 &= 10. \end{aligned}$$

- (a) Write the system in matrix form $\mathbf{Ax} = \mathbf{b}$.
- (b) Compute the determinant of the coefficient matrix.
- (c) Are the two equations linearly independent?
- (d) Does the system have a unique solution?
- (e) Explain economically why these two restrictions do not really provide two independent pieces of information.
- (f) Give one different second equation that would make the coefficient matrix nonsingular.

11. **Based on class discussion.** In many economic models, we want to know whether a set of basic activities can generate all possible changes in two variables. For example, suppose an economy has two production activities. Activity 1 changes the output vector by

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix},$$

and activity 2 changes the output vector by

$$\mathbf{v}_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

Any combination of these two activities produces a vector of the form

$$a\mathbf{v}_1 + b\mathbf{v}_2,$$

where $a, b \in \mathbb{R}$.

- (a) Write $a\mathbf{v}_1 + b\mathbf{v}_2$ as a vector in terms of a and b .
- (b) Can the activities generate the vector

$$\begin{bmatrix} 3 \\ 4 \end{bmatrix}?$$

If yes, find values of a and b .

- (c) Can the activities generate the vector

$$\begin{bmatrix} 1 \\ 7 \end{bmatrix}?$$

If yes, find values of a and b .

- (d) Use the determinant of the matrix

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}$$

to determine whether $\mathbf{v}_1$ and $\mathbf{v}_2$ are linearly independent.

- (e) Do $\mathbf{v}_1$ and $\mathbf{v}_2$ span $\mathbb{R}^2$? Explain.

- (f) Give an economic interpretation of what it means for these two activity vectors to span $\mathbb{R}^2$.

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